Multimodal fusion improves fire detection accuracy compared to single-modality models.

RGB + Thermal Fusion

Multimodal Wildfire Detection

Problem & Motivation

Rapid wildfire detection is critical

- Single sensors produce unreliable predictions
- Fusion improves accuracy

Model Training

- YOLOv11 fine-tuned on datasets
- RGB & Thermal models trained independently
- Higher resolution

System Architecture

- RGB & thermal process separately
- Models generate fire probabilities
- Fusion produces final detection decision

Fusion Methods Tested

- Weighted fusion adds probabilities
- Product fusion for sensor agreemen
- Methods evaluated using accuracy and F1 score

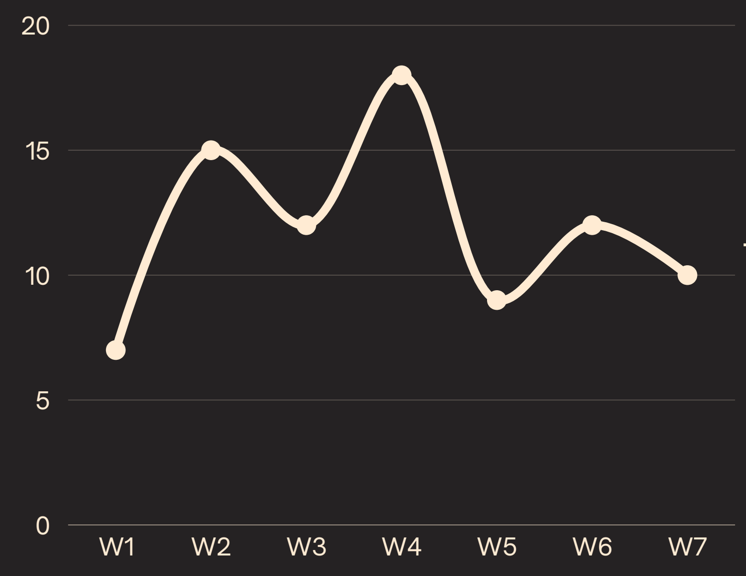
Project Completion

68% Complete

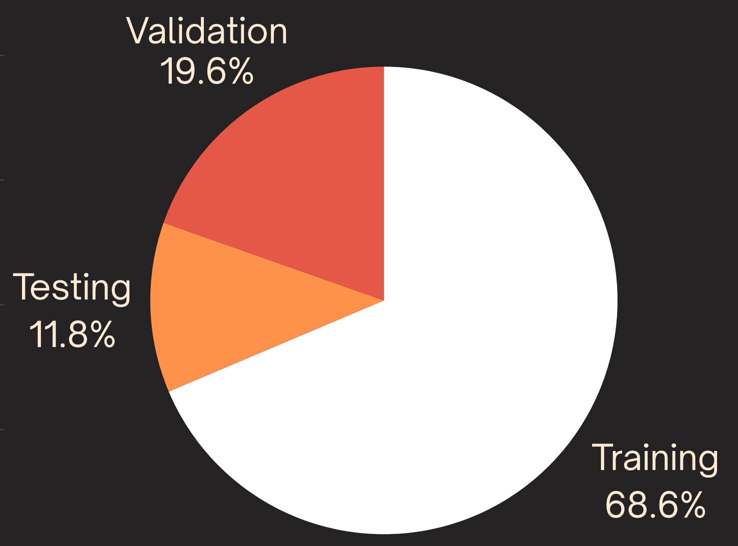
Model Accuracy

85% Accuracy

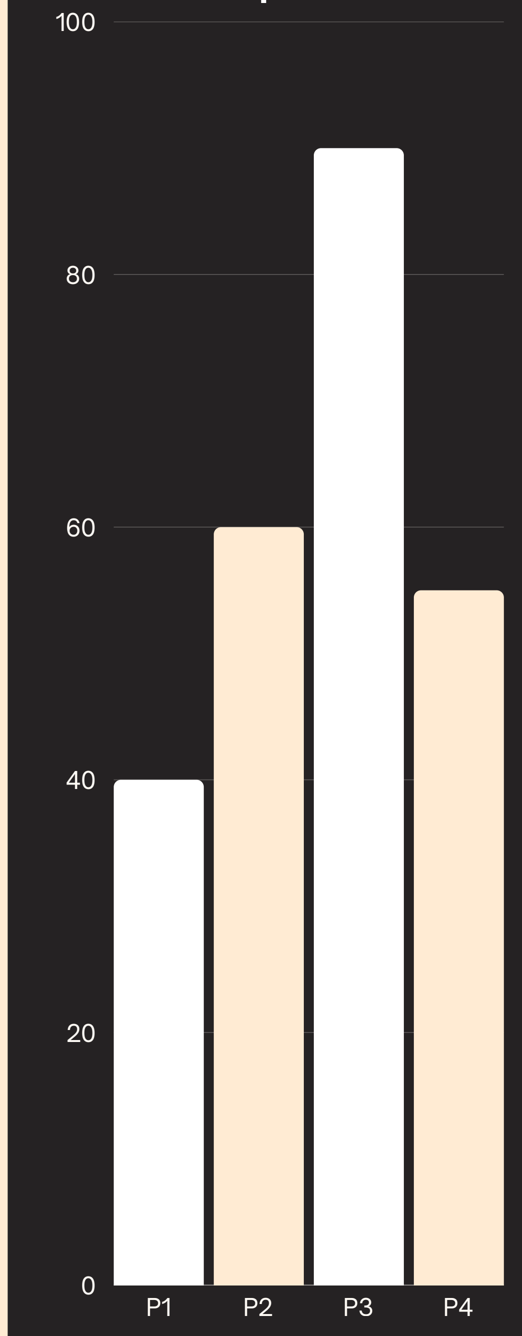
Training Loss vs Epoch



Dataset Distribution



Fusion Method Comparison



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